AICTE – BharatCares – IBM SkillsBuild Internship 2026

PROJECT CONCEPT NOTE

Machine Learning & Applied AI Track | Final Capstone Submission

Project Title:	PHISHGUARD//NEXUS: Explainable AI Phishing & Scam Detection Platform
Student Name:	Sahil Rai
Internship ID:	IBMUEML002432
Academic Institution:	Lovely Professional University (LPU), Punjab
Live Cloud Deployment:	https://phishguard-nexus.vercel.app/ (Shortlink: tinyurl.com/pg-nexus)
GitHub Repository:	https://github.com/rootbysahil/phishguard-nexus

1. Executive Summary

Digital communications in India face an exponential rise in credential theft and fake KYC scams. Conventional defenses rely on static domain blacklists and opaque black-box neural networks. **PHISHGUARD//NEXUS** resolves both issues through an Explainable AI (XAI) platform that evaluates 24 lexical signals without network queries (<15ms latency), achieving an ROC-AUC of 1.000 and F2 of 0.9989 with calibrated Logistic Regression. Localized SHAP feature attributions and IBM watsonx / Granite LLM multilingual advisories directly empower citizens and route victims to the National Cybercrime Portal (Helpline 1930 / cybercrime.gov.in).

2. UN Sustainable Development Goals (SDGs)

• **SDG 9 (Industry, Innovation & Infrastructure - Target 9.c):** Delivers zero-latency, privacy-first cybersecurity infrastructure accessible to Indian citizens, educational bodies, and MSMEs.
• **SDG 16 (Peace, Justice & Strong Institutions - Target 16.4 & 16.5):** Directly tackles digital financial fraud, illicit money diversion, and extortion.

3. Technical Architecture & Machine Learning Pipeline

• **Feature Engineering (24 Pure-Python Signals):** Shannon entropy, Punycode/homoglyph detection, brand keyword displacement in subdomains, path depth, TLD risk analysis. Zero external network queries eliminate victim IP exposure.
• **Model Benchmarking:** Evaluated across Heuristic Baselines, Decision Trees, Random Forests, and Platt-Calibrated Logistic Regression. Calibrated LR achieved optimal generalization with ROC-AUC 1.000, F2 0.9989, and Recall 1.000.
• **Leakage Prevention:** Strict GroupShuffleSplit on registered domain families; zero domain overlap between train and test splits.
• **Explainable AI (XAI):** LinearExplainer assigns directional weights to positive and negative indicators.
• **IBM watsonx.ai Layer:** Orchestrates plain-language translation in English, Hindi, and Punjabi.
• **Cloud Deployment:** Serverless Python ASGI on Vercel with automated ReportLab PDF generation.

4. SkillsBuild 6-Week Masterclass Milestones

Week	Masterclass Module	Project Accomplishment Executed
Week 1	Kickstart ML & Tools (VS Code, BOB)	Project setup, dataset acquisition, architectural contracts.
Week 2	Supervised Learning & IBM watsonx	24-feature extractor, classifier benchmarking & calibration.
Week 3	Unsupervised Patterns & Deep Learning	Entropy analysis, homoglyph clustering, GroupShuffle validation.
Week 4	Intelligent Agents & Lifelong Skills	SHAP XAI engine, watsonx advisory prompt engineering.
Week 5	Project Completion & Q&A; Session	FastAPI serverless backend, Cyberpunk UI & ReportLab PDF engine.
Week 6	Project Submission (PPT & Code)	Live Vercel cloud deployment, automated test verification (8/8).

5. Formal Submission Undertaking

I hereby declare that this project, **PHISHGUARD//NEXUS**, has been developed as my final capstone deliverable for the 6-week virtual IBM SkillsBuild Machine Learning and Applied AI Internship (2026) in collaboration with AICTE and BharatCares. The system is verified, passing all unit and integration tests (8/8), and live at **https://phishguard-nexus.vercel.app/**.

Student Name: Sahil Rai **Internship ID:** IBMUEML002432 **Date:** 13/10/2026